

A Grammar of Physical Theories

Structure, constitution and interaction in physical models

Paper I of IV

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Abstract

Physical theories possess a modular architecture that formula-sheet presentation largely conceals. Following Tonti's classification of configuration-kind and source-kind variables [1, 2], this essay distinguishes equations by the role they play within a theory: structure equations construct derived quantities, constitutive equations connect different descriptions, dynamical equations govern change, and interaction equations couple theoretical sectors. The framework is developed through classical and relativistic particle mechanics, Newtonian gravity, electrostatics and magnetostatics. It also explains a recurring pedagogical phenomenon: many familiar equations function as *hinges*, preserving one module of the theoretical grammar while leaving another slot open. A companion essay, *The Anatomy of an Equation* (Paper III), introduces a second, orthogonal classification separating an equation's architectural role from its epistemic status. (Paper IV) addresses the modeling choices.

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From formula sheets to theoretical architecture

Physics is often presented as an inventory of equations: velocity is the derivative of position, momentum is mass times velocity, force is the rate of change of momentum, electric field is the gradient of potential, and charge density is the divergence of electric displacement. The sequence conceals the architecture that makes the statements parts of a theory rather than entries on a formula sheet. A structural question to ask prior to calculation is thus:

What kinds of quantities does the theory contain, and what roles do its equations play?

This question has a direct genealogy in the work of Enzo Tonti, whose 1975 monograph proposed a classification of physical variables into configuration-kind and source-kind families, connected by constitutive equations and embedded within a wider system of structural and interaction relations [1]. Figure 1 reconstructs his preliminary classification. A later, more developed presentation appears in his 2013 book [2]. The present essay uses more elementary grammar because it exposes the conceptual roles of equations before the additional topological machinery is introduced.

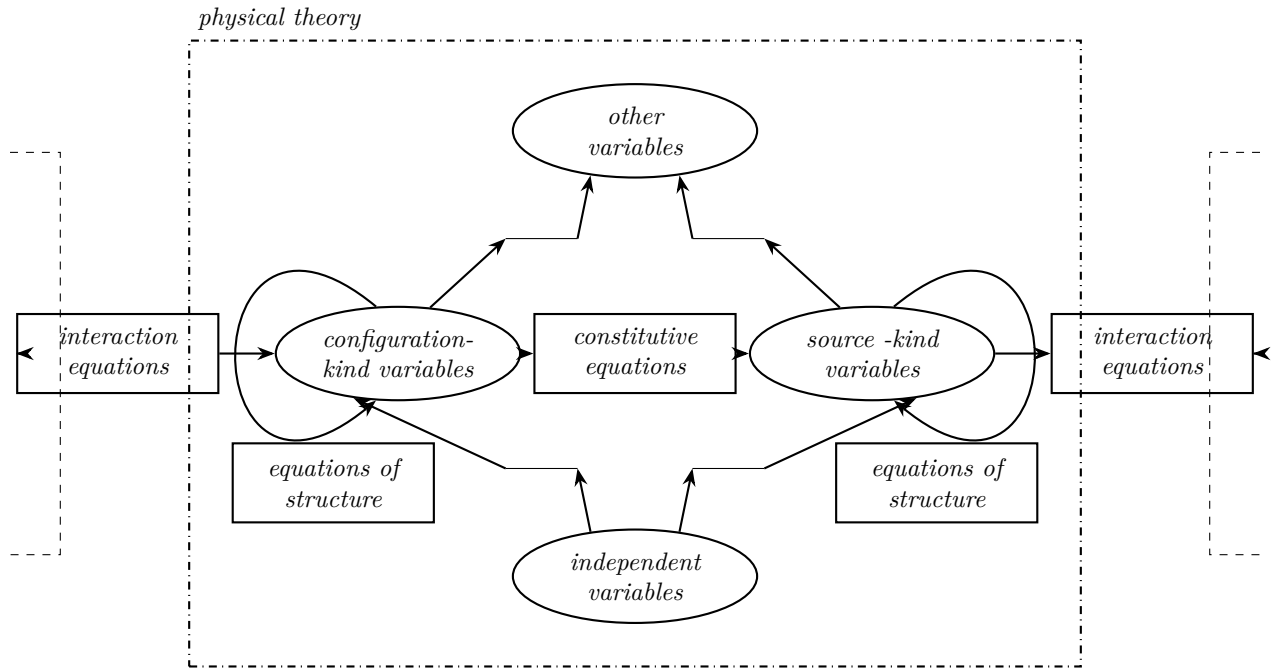


Figure 1: A general grammar for the variables and equations of a physical theory, adapted from Tonti's classification [1]. The same mathematical object may occupy different roles in different formulations.

The claim here is modest in form but substantial in consequence. A theory is more than a set of propositions; it is an organised network in which quantities and equations occupy different functional positions. Once those positions are visible, one can ask precisely what changes when a theory is modified, what remains invariant, and which statement is still required before a model becomes physically determinate.

The modularity exposed here also bears directly on a persistent finding in physics education research: that students sort problems by surface features while experts sort them by underlying physical principle [3]. Sherin's account of *symbolic forms* describes expert equation-reading as the simultaneous activation of a notational pattern and a conceptual schema [4]. Redish characterises mathematical use in physics as a disciplinary dialect in which form and physical meaning are continuously blended [5]. The present framework offers one account of what those experts are recognising: the architectural role an equation plays, and the slot it leaves open for case-specific completion.

The variable grammar of a physical theory

The diagram should not be read as a single causal chain. Some arrows are definitions, some are compatibility conditions, some are dynamical maps, and some are transformations of representation. The point is structural: a theory contains two principal families of variables, equations internal to each family, equations translating between them, and couplings across its boundary.

Three cautions on reading this. First, *source* does not always mean an externally imposed cause. Momentum is source-kind in the mechanics diagram because its rate of change is force; charge density is source-kind because it appears in a field equation. Second, *constitutive laws* need not describe matter alone as in the map between velocity and momentum is constitutive in mechanics. Third, the classification depends upon formulation. So in a Lagrangian description, velocity is generated from configuration; in a Hamiltonian description, velocity is recovered from momentum through a derivative of the Hamiltonian.

The upper node also requires interpretation. Energy is a natural example of a derived or composite variable because it can be assembled from both sides:

$$H(\mathbf{r}, \mathbf{p}) = T^*(\mathbf{p}) + V(\mathbf{r}).$$

Power similarly combines source-kind and configuration-rate variables through $P = \mathbf{F} \cdot \mathbf{v}$. These quantities do not disappear from the theory, rather they occupy a different functional position from the primary configuration and source families.

A useful preliminary classification is shown in Table 1.

Element	Role in the theory
Independent variables	Coordinates or parameters with respect to which the theory is formulated, such as time t , position $\mathbf{x}$, or spacetime coordinates x^μ .
Configuration-kind variables	Variables that specify a state, trajectory, potential or geometric configuration.
Source-kind variables	Variables that encode momentum, force, density, current, stress or another source, response or conjugate quantity.
Structure equations	Kinematic or geometric relations that construct derived quantities or impose compatibility within one family of variables.
Constitutive equations	Relations connecting configuration-kind and source-kind descriptions. They frequently carry material, regime-dependent or theory-specific content.
Dynamical or balance equations	Relations governing change, transport or conservation, such as $\dot{\mathbf{p}} = \mathbf{F}$ or a divergence balance.
Interaction equations	Couplings through which one theoretical sector supplies a force, source or response to another.
Derived or composite variables	Quantities assembled from both sides of the theory, such as energy, power or action.

Table 1: Functional elements in a physical theory whose assignments are formulation-dependent.

Three kinds of theoretical completion

The distinctions that will organise this paper are between a *structural relation*, a *constitutive closure*, and an *interaction coupling*.

A *structure equation* constructs or constrains quantities within one side of the theory:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt}, \quad \mathbf{E} = -\nabla\phi, \quad \mathbf{B} = \nabla \times \mathbf{A}.$$

A *constitutive equation* translates between different descriptions:

$$\mathbf{p} = m\mathbf{v}, \quad \mathbf{D} = \varepsilon\mathbf{E}, \quad \mathbf{B} = \mu\mathbf{H}.$$

It carries information that may change with material, physical regime or theoretical framework.

An *interaction equation* couples one sector to another:

$$\mathbf{F} = m_g\mathbf{g}, \quad \mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad \rho_q(\mathbf{x}, t) = q\delta(\mathbf{x} - \mathbf{r}(t)).$$

These categories are not interchangeable. A structure relation may leave the mechanism unspecified while a constitutive relation may close one translation while leaving the external interaction open. Ineed an interaction law may provide a force while relying on a separately specified field. A complete physical account normally requires several such modules, together with a choice of boundary, approximation and independent variables.

This gives us an initial formulation of the hinge idea developed and implemented in a companion pedagogical essay [10]:

A hinge relation exposes an open theoretical slot. Depending on the case, that slot is completed by a constitutive law, an interaction coupling, a boundary condition, a regime assumption, or a specification of the system.

The point is not that every hinge is the same kind of equation. Rather the hinges is one part of the architecture that can be held fixed while another is deliberately varied.

Classical particle mechanics: where the programme becomes visible

For a particle, the independent variable is time t , and position $\mathbf{r}(t)$ is the primary configuration variable. Classical mechanics may then be written as a loop:

$$\mathbf{r} \xrightarrow{\text{kinematics}} \mathbf{v} \xrightarrow{\text{constitution}} \mathbf{p} \xrightarrow{\text{dynamics}} \mathbf{F} \xrightarrow{\text{interaction}} \mathbf{r}.$$

The loop contains four distinct theoretical acts:

1. The kinematic relation $\mathbf{v} = \dot{\mathbf{r}}$ constructs velocity from trajectory.
2. The constitutive relation $\mathbf{p} = m\mathbf{v}$ connects kinematic velocity to momentum.
3. The dynamical balance $\dot{\mathbf{p}} = \mathbf{F}$ relates momentum change to force.
4. The interaction law $\mathbf{F} = \mathbf{F}(\mathbf{r}, \mathbf{v}, t, \dots)$ specifies the mechanism producing the force and thereby closes the trajectory problem.

Figure 2 below is the rigorous version of the *placeholder force* argument developed in the teaching companion essay [9].

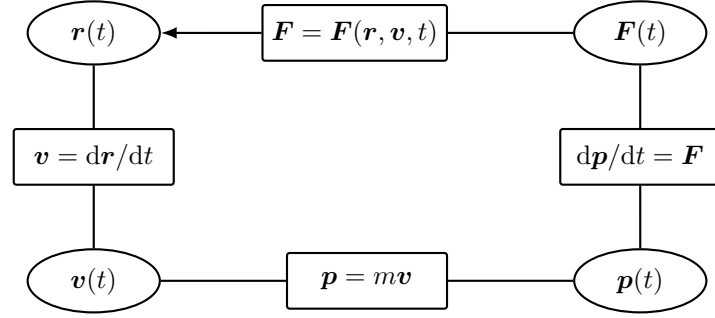


Figure 2: Classical particle mechanics as a closed loop. The trajectory is not determined until an interaction law supplies the force.

The familiar equation $\mathbf{F}_{\text{net}} = m\mathbf{a}$ is dynamically operative while remaining physically uninstantiated and the compression

$$\mathbf{v} = \dot{\mathbf{r}}, \quad \mathbf{p} = m\mathbf{v}, \quad \dot{\mathbf{p}} = \mathbf{F} \implies \mathbf{F} = m\mathbf{a}$$

is efficient only once the full architecture is understood. Before that point, it conceals the distinction between definition, constitution and dynamics. Physics education research has documented the consequences of this concealment. Clement’s reversal-error studies showed that students read algebraic symbols as labels for nouns rather than as variables denoting measured quantities [6]. Torigoe’s work on *unpacking* demonstrated that successful interpretation requires students to recover the physical commitments compressed into the notation [7]. The present framework explains why such unpacking is structurally necessary: the compressed equation fuses three modules that occupy distinct architectural positions.

The hinge in context

The mechanics loop above explains why certain equations work naturally as pedagogical hinges. A hinge is a relation held fixed across several cases while students identify the additional statement required to instantiate the model. In the present terminology, the hinge is commonly a structure, balance or dynamical relation, while the case-specific completion is commonly constitutive, interactional, geometric or conditional. For example,

$$\frac{d\mathbf{p}}{dt} = \mathbf{F}$$

organises the dynamical relation between momentum change and force but it does not specify the physical origin of $\mathbf{F}$. The cases

$$\mathbf{F} = m\mathbf{g}, \quad \mathbf{F} = -k\mathbf{x}, \quad \mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

supply distinct interaction closures. Likewise, the wave relation $v = f\lambda$ may be completed by a medium model such as $v = \sqrt{T/\mu}$ or $v = 1/\sqrt{\epsilon\mu}$. The classroom comparison is structurally legitimate because one module is held invariant while another is varied.¹

Kuo and colleagues showed that successful physics problem-solving blends conceptual reasoning with formal mathematical manipulation rather than alternating between the two [8]. The hinge

¹The companion hinge programme develops this pedagogical consequence in detail: *Teaching Through the Hinge* [10].

framework provides a structural account of what is being blended: the invariant module (structure, dynamics) provides the formal scaffold, while the case-specific module (constitution, interaction) provides the physical reasoning. Students who have learned equations primarily as procedures for finding x are being asked, in Redish's terms, to enter a different semiotic practice [5]. The grammar makes explicit what that practice requires.

Relativistic mechanics: the hinge survives a change of theory

Special relativity provides the clearest example of a local modification within the architecture. In the three-vector formulation, the kinematic and dynamical relations retain their form:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt}, \quad \frac{d\mathbf{p}}{dt} = \mathbf{F}.$$

The decisive change is the constitutive map between velocity and momentum. Classical mechanics uses $\mathbf{p} = m\mathbf{v}$; relativistic mechanics uses

$$\mathbf{p} = \gamma m_0 \mathbf{v}, \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

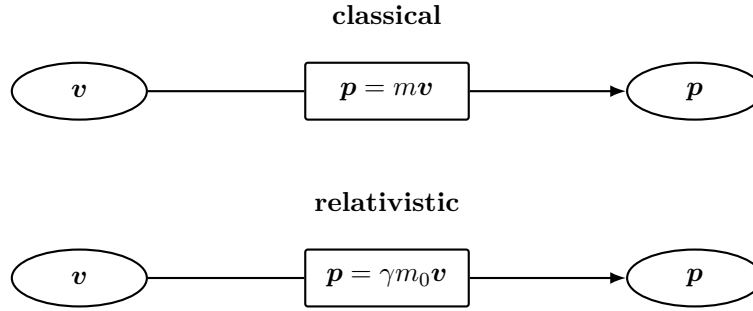


Figure 3: Classical and relativistic mechanics share the same velocity–momentum slot while supplying different constitutive maps.

The educational lesson is precise: a dynamical hinge can survive a change of theory while its constitutive closure changes. This is stronger than saying that relativity supplies a different formula in identifying *where* the theoretical modification occurs.

The spacetime formulation makes the continuity still clearer. With proper time τ ,

$$u^\mu = \frac{dx^\mu}{d\tau}, \quad p^\mu = m_0 u^\mu, \quad \frac{dp^\mu}{d\tau} = K^\mu.$$

Position, four-velocity, four-momentum and four-force occupy the same successive roles as their Newtonian counterparts, now constrained by Minkowski geometry. Relativity therefore appears as a controlled reorganisation of the same grammar rather than an unrelated inventory.

Gravity and electrostatics: parallel field grammars

Newtonian gravity and electrostatics share a striking static architecture. Each begins with a scalar potential, differentiates it to obtain a field, relates that field to a flux-like variable, and takes a divergence to recover a source density.

For gravity², define

$$\mathbf{g} = -\nabla V, \quad \mathbf{h} = -\frac{\mathbf{g}}{4\pi G}, \quad \nabla \cdot \mathbf{h} = \rho_m.$$

Combining the equations gives $\nabla^2 V = 4\pi G \rho_m$. A test body couples to the gravitational field through $\mathbf{F} = m_g \mathbf{g}$, yielding $m_i \ddot{\mathbf{r}} = m_g \mathbf{g}$. The equivalence $m_i = m_g$ removes the test body's mass from its acceleration.

For electrostatics, $\mathbf{E} = -\nabla \phi$, $\mathbf{D} = \varepsilon \mathbf{E}$, $\nabla \cdot \mathbf{D} = \rho_q$, and the particle-field interaction is $\mathbf{F} = q \mathbf{E}$.

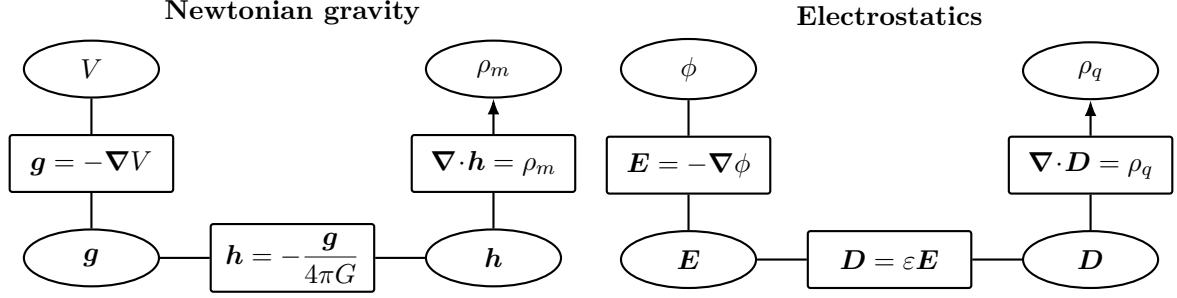


Figure 4: Parallel static-field grammars. The sequence potential $\rightarrow$ field $\rightarrow$ flux variable $\rightarrow$ source is preserved while signs, constants, source types and interaction laws differ.

The analogy is formal rather than ontological. Mass has one observed sign and Newtonian gravity is attractive; electric charge has two signs and the electrical interaction may attract or repel. Yet the common architecture reveals where the theories agree and where they diverge.

Moving charge and the electromagnetic feedback loop

Magnetostatics introduces a vector potential, $\mathbf{B} = \nabla \times \mathbf{A}$. For a linear isotropic medium, $\mathbf{B} = \mu \mathbf{H}$, and in the magnetostatic limit Ampère's law is $\nabla \times \mathbf{H} = \mathbf{J}$. For a continuous moving charge distribution, $\mathbf{J} = \rho_q \mathbf{v}$. The source is generated from both charge density and motion, which is the map joining particle kinematics to magnetic field theory.

All that is beyond High school. What is not (apparently) is that the feedback to particle mechanics is provided by the Lorentz interaction:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

A self-consistent particle-field architecture therefore has the form

$$(\mathbf{r}, \mathbf{v}) \longrightarrow (\rho_q, \mathbf{J}) \longrightarrow (\phi, \mathbf{A}) \longrightarrow (\mathbf{E}, \mathbf{B}) \longrightarrow \mathbf{F} \longrightarrow (\mathbf{r}, \mathbf{v}).$$

Interaction equations inside and outside the theory boundary

The original Tonti diagram places interaction equations at the boundary because it represents the coupling of one theoretical sector to another. Figure 5, overleaf by contrast, appears to internalise the interaction. There is no contradiction as the location of the interaction depends upon the chosen level of closure.

For a particle in a prescribed external field, $\mathbf{E}$ and $\mathbf{B}$ belong to an external sector. The Lorentz force is a boundary interaction, and the particle's effect on the field is omitted. This is a legitimate

²The symbol $\mathbf{h}$ is introduced here solely to display the Newtonian equations in a flux–source form parallel to electrostatics; it is not intended as a standard gravitoelectromagnetic field variable.

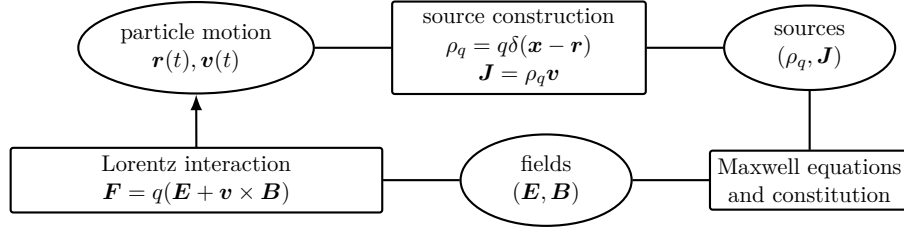


Figure 5: A closed particle–electromagnetic-field coupling. Source generation and back-reaction are both internal to the enlarged system.

test-particle approximation. For a self-consistent particle-field model, the particle and field are included inside one enlarged system and the source map from motion to $(\rho_q, \mathbf{J})$ and the back-reaction map from $(\mathbf{E}, \mathbf{B})$ to force are both internal.

Whether an interaction equation appears inside or outside the theory boundary depends on the chosen level of closure. A prescribed-field model treats the field as external; a self-consistent particle-field model enlarges the theory until source generation and back-reaction are internal.

This is why an apparently incomplete diagram may represent a controlled approximation rather than an error. What matters is that the omitted feedback and the chosen boundary are made explicit.

What the grammar clarifies

The framework makes comparison between theories precise.

	Configuration or structural side	Constitutive map	Source, balance or interaction side
Classical mechanics	$\mathbf{r} \mapsto \mathbf{v}$	$\mathbf{p} = m\mathbf{v}$	$\dot{\mathbf{p}} = \mathbf{F}$, plus a force law
Relativistic mechanics	$x^\mu \mapsto u^\mu$	$p^\mu = m_0 u^\mu$, $\mathbf{p} = \gamma m_0 \mathbf{v}$	$dp^\mu/d\tau = K^\mu$
Newtonian gravity	$V \mapsto \mathbf{g}$	$\mathbf{h} = -\mathbf{g}/(4\pi G)$	$\nabla \cdot \mathbf{h} = \rho_m$, $\mathbf{F} = m_g \mathbf{g}$
Electrostatics	$\phi \mapsto \mathbf{E}$	$\mathbf{D} = \varepsilon \mathbf{E}$	$\nabla \cdot \mathbf{D} = \rho_q$, $\mathbf{F} = q\mathbf{E}$
Magnetostatics	$\mathbf{A} \mapsto \mathbf{B}$	$\mathbf{B} = \mu \mathbf{H}$	$\nabla \times \mathbf{H} = \mathbf{J}$

Table 2: Different theories as variations within a common grammar.

Several conclusions follow. A change of theory may be local within the architecture: the classical-to-relativistic transition modifies one constitutive cell while preserving the others. Formally parallel theories may differ in source type, sign structure and interaction law. Familiar equations often compress several modules: $\mathbf{F} = m\mathbf{a}$ fuses kinematics, constitution and dynamics; $\nabla^2 \phi = -\rho_q/\varepsilon$ fuses structure, constitution and source. Compression is efficient after the architecture is understood; before then, it encourages the treatment of equations as isolated prescriptions.

Toward a second axis: the epistemic status of equations

Tonti’s framework classifies an equation by its *architectural role*: where it sits within a theory and which variable families it connects. A second question remains:

What kind of claim does the equation make?

Consider $\nabla \cdot \mathbf{B} = 0$. If $\mathbf{B} = \nabla \times \mathbf{A}$, then $\nabla \cdot \mathbf{B} = \nabla \cdot (\nabla \times \mathbf{A}) = 0$ by a vector-calculus identity. Yet the physical availability of a single local vector potential rests on the absence of magnetic charge. If magnetic monopoles were included, the equation would become $\nabla \cdot \mathbf{B} = \rho_m^{\text{mag}}$, and the same representation would fail. The equation is simultaneously a mathematical identity (given the potential representation) and a physical law (asserting the non-existence of magnetic charge). Its architectural role, structural constraint, does not settle its epistemic status.

This observation points toward a second, orthogonal classification: whether an equation is a definition, an identity, an empirical law, a constitutive assumption, an approximation, an idealisation, a boundary condition, or a derived result. The companion essay *The Anatomy of an Equation* (Paper III) develops this classification systematically, constructing a matrix in which architectural role and epistemic status form independent axes and examining equations whose status changes under a change of formulation or a change of physical theory.

Conclusion

The modular architecture of a physical theory is often hidden by its most efficient formulas. Tonti's classification restores the separation between configuration-kind and source-kind variables and identifies the structural, constitutive and interactional relations joining them. Classical mechanics makes the point vivid: trajectory generates velocity, constitution generates momentum, dynamics relates momentum change to force, and an interaction law closes the loop. The force symbol is therefore a placeholder with a determinate theoretical role.

Relativity shows that a new theory need not replace the whole architecture; the dynamical hinge survives while its constitutive closure changes. Gravity and electrostatics show how a common field grammar can be preserved while sources, signs and interactions differ. Electromagnetism exposes the dependence of theoretical closure on the placement of the system boundary.

The pedagogical consequence follows directly. A hinge relation preserves one module of the theoretical grammar while exposing another as open. The missing completion may be constitutive, interactional, geometric, conditional or systemic. The classroom experience of needing "something else" is the visible residue of the modular structure of physical theory.

Very few educational papers discuss where the system boundary is drawn. But a theory is more than a collection of equations, rather a declaration of what lies inside the model and what crosses its boundary.

This Paper I has argued that equations occupy positions within an architecture before they possess epistemic status. Understanding the grammar of a theory is therefore logically prior to asking what kind of claim an individual equation makes. That second question motivates Paper III.

References

- [1] E. Tonti, "On the formal structure of physical theories," *Quaderni dei Gruppi di Ricerca Matematica del CNR*, Istituto di Matematica del Politecnico di Milano, 1975.
- [2] E. Tonti, *The Mathematical Structure of Classical and Relativistic Physics: A General Classification Diagram*, Modeling and Simulation in Science, Engineering and Technology, New York: Birkhäuser/Springer, 2013. doi: 10.1007/978-1-4614-7422-7.
- [3] M. T. H. Chi, P. J. Feltovich, and R. Glaser, "Categorization and representation of physics problems by experts and novices," *Cognitive Science*, vol. 5, no. 2, pp. 121–152, 1981. doi: 10.1207/s15516709cog0502_2.

- [4] B. L. Sherin, “How students understand physics equations,” *Cognition and Instruction*, vol. 19, no. 4, pp. 479–541, 2001. doi: 10.1207/S1532690XCI1904_3.
- [5] E. F. Redish, “Problem solving and the use of math in physics courses,” in *Proceedings of the Conference World View on Physics Education in 2005: Focusing on Change*, Delhi, India, pp. 1–10, 2005. Available: arXiv:physics/0608268.
- [6] J. Clement, “Algebra word problem solutions: Thought processes underlying a common misconception,” *Journal for Research in Mathematics Education*, vol. 13, no. 1, pp. 16–30, 1982. doi: 10.2307/748434.
- [7] E. T. Torigoe, “Unpacking symbolic equations in introductory physics,” in *Physics Education Research Conference 2015*, College Park, MD: American Association of Physics Teachers, pp. 327–330, 2015. Available: arXiv:physics/1508.00535.
- [8] E. Kuo, M. M. Hull, A. Gupta, and A. Elby, “How students blend conceptual and formal mathematical reasoning in solving physics problems,” *Science Education*, vol. 97, no. 1, pp. 32–57, 2013. doi: 10.1002/sce.21043.
- [9] L. McCulloch-James, “The placeholder force: How the same F conceals different physics,” 2025. Available: leelemacjames.github.io.
- [10] L. McCulloch-James, “Teaching through the hinge,” 2025. Available: leelemacjames.github.io.